North Star: Goals, Expectations & Focus

Framework Goals

What the framework is built to achieve. Each goal traces back to a section on this board.

1	**Measure outcomes, not activity**	Tie every metric to an OKR; prove Net Time Saved, not usage
2	**Govern every agent**	100% of agents in the Agent Catalog; no Shadow AI
3	**Earn autonomy safely**	High agency inside guardrails; HITL → HOTL as evidence builds
4	**Protect quality, shrink the Verification Tax**	Raise first-pass acceptance; keep rework and AI defects down
5	**Reuse before rebuild**	One shared platform, N value-stream instances; durable skills
6	**Meet in the middle**	Leadership metrics and team adoption converge through middle management
7	**Learn continuously**	Self-improving loop: adopt → measure → learn → codify → scale
8	**Operate at scale**	Reliably and cost-efficiently, within SLOs and token budgets

Traces to: 1 OKR → Telemetry Cascade · 2 Agent Catalog / Phased Rollout Stage 1 · 3 Agent Control spectrum, tier autonomy calibration · 4 Verification Tax, Dimensions (Quality) · 5 Shared Capability Map · 6 Phased Rollout (meet in the middle) · 7 At Scale learning loop · 8 SLI/SLO/SLA breakdown, Dimensions (Economics)

Telemetry First Platform Solution

Priority #1: set up end-to-end logging before building out the foundational agent platform

You can't govern, improve or prove the ROI of what you can't see. If telemetry isn't in place first, there are no baselines, no Verification Tax, no error budgets, and every later decision is argued from opinion instead of evidence.

1) START HERE
INSTRUMENT: End-to-end logging across prompt → agent → tool → output → review → merge. One trace ID per task.

2) BASELINE
Capture pre-AI workflow time, quality and cost, so Net Time Saved and Verification Tax have a starting point.

3) BUILD THE PLATFORM
Agent Catalog, harness, guardrails and shared capabilities, built on top of live telemetry.

4) SCALE WITH EVIDENCE
Widen autonomy and onboard new waves only when SLOs and error budgets show it's safe.

What "end-to-end" must capture
- **Who / what:** user, team, role, agent ID and version, model, skill invoked
- **Every step:** prompt, context retrieved, tool and MCP calls, outputs, latency, tokens and cost
- **Human touchpoints:** accept / edit / reject, overrides, HITL approvals, review time
- **Outcomes:** build and test result, PR merged, defects traced to AI, rollbacks
- **Safety events:** policy and secret violations, guardrail blocks
- **Correlation:** one trace ID linking all of it, retained per data policy

Exit criteria before the platform build-out
- Nearly all agent runs emit a complete trace (e.g. ⩾ 95%)
- Dashboards live for the Dimensions SLIs, by lens (L / M / IC)
- Baselines captured for every in-scope workflow
- Verification Tax and Net Time Saved computable straight from logs
- PII / secret redaction and retention policy approved
- Alerts wired to SLOs and error budgets

⚠ **Anti-pattern:** launching agents first and "adding observability later." The pre-AI baseline is lost for good, the Delivery J-curve dip looks like failure, and Shadow AI Agents goes unseen.

Maturity Model + Loop Engineering

Purpose: show how each agent moves up the maturity levels as it moves through its development lifecycle. The goal is Level 6 (fully agentic, AI-native), but every agent climbs one level at a time and earns each step with evidence. Risky agents may stop in the middle tiers and keep a human in the loop. Runs on the continuous **Assess → Prioritize → Execute → Measure → Adapt** 90-day transformation loop.

How an agent climbs: one level at a time, each step earned with evidence

▼ High-risk ceiling (stays HITL) · ▼ Medium-risk ceiling (HOTL) · ▼ Target: low-risk agents

0 · Experimental
AI exploration
Human does the work
→
1 · Assisted
Copilots
Human drives every step; AI suggests
→
2 · Standardized
Common platform
Human approves every output
→
3 · Governed
Controlled agents
Human approves each action (HITL)
→
4 · Agentic
Multi-step autonomy
Human supervises, steps in on alerts
→
5 · Adaptive
Dynamic systems
Human sets policy; agent self-evaluates
→
6 · Autonomous Enterprise
Outcome-driven AI
Humans govern the bounds and audit

Oversight: no loop · human does or drives the work

Open loop · Human-in-the-loop (HITL)
active human intervention; human review guards against comprehension debt

Closed loop · Human-on-the-loop (HOTL)
verification pipeline + feedback telemetry (SQY); supervisor governs from outside

Closed loop · Human-out-of-the-loop
sealed, self-correcting

Lifecycle: 1 Design & Baseline · 2 Offline Evaluation · 3 Pilot (HITL) · 4 Production (HOTL) · Production, steady state

*At every level: **5 Periodic Review** (AI Yield per agent, SLOs, error budget) → **6 Decide**: promote one level · keep · demote · new version vN+1 (re-enters at Design & Baseline) · retire. Stages match **Agent Lifecycle Evaluation**; loop modes match **HITL vs HOTL: Open Loop vs Closed Loop**.*

▲ Promotion gates: move up one level only when
- Offline eval meets the baseline on the agent's eval set (Offline Human Alignment & Analysis)
- SLOs held for a full review period with the error budget intact
- The verification pipeline's staged checks catch failures before output; human review stops finding new failure modes
- AI Yield > 1.0 at Periodic Review
- Telemetry and audit trail complete for every action

No skipping levels: each one produces the evidence the next one needs.

■ Risk ceiling: where an agent stops
- **High risk** (regulated, financial, safety, irreversible actions): ceiling Level 3 Governed. Stays open loop; a human approves every action (HITL)
- **Medium risk** (customer-facing, reversible): ceiling Level 4–5. Closed loop with a human on the loop (HOTL); the supervisor is alerted on SLI breach
- **Low risk** (internal, easily reversed): may reach Level 6, human out of the loop within bounded governance

A ceiling is a design choice, not a failure: a Level 3 agent with high AI Yield is a success.

▼ Demotion: when an agent moves down
- SLO breach or error budget exhausted → drop one level and restore the human checkpoint
- Comprehension debt rising (people can no longer explain what the agent did) → back to HITL
- Incident or policy violation → back to Level 3; review before re-promotion
- Major change to model, prompts or tools → new version vN+1 re-enters at Design & Baseline

Maturity levels in detail

Level	State	Characteristics	Human role · loop mode	Lifecycle stage	Promote to next level when...
0 — Experimental	AI exploration	Individual tools, no governance	Human does the work · no loop	1 Design & Baseline	Use case, owner and baseline metrics defined
1 — Assisted	Copilots	Human-driven, fragmented telemetry	Human drives every step; AI suggests · no loop	1 Design & Baseline	Telemetry captured end to end on the common platform
2 — Standardized	Common platform	Approved models, tools, standards	Human approves every output · open loop (HITL)	2 Offline Evaluation	Offline eval meets baseline; human alignment labels in place
3 — Governed	Controlled agents	Policy-as-code, HITL, audit	Human approves each action · open loop (HITL). Ceiling for high-risk agents	3 Pilot (HITL)	Pilot SLOs met; human review finds no new failure modes; AI Yield > 1.0
4 — Agentic	Multi-step autonomy	Workflows, DAGs, memory, tools	Human supervises, steps in on alerts · closed loop (HOTL)	4 Production (HOTL)	Verification pipeline and supervisor alerts hold SLOs for a full review period
5 — Adaptive	Dynamic systems	Routing, optimization, self-evaluation	Human sets policy; agent self-evaluates · closed loop (HOTL). Ceiling for medium-risk agents	4 Production (HOTL)	Self-evaluation agrees with human audits; error budget intact
6 — Autonomous Enterprise	Outcome-driven AI	Closed-loop execution with bounded governance	Human out of the loop; humans govern bounds and audit · sealed closed loop	Production, steady state	Top level: stays while AI Yield > 1.0 and SLOs hold; otherwise demote

From Pilot to Platform: Scaling Adoption Through Continuous Learning

Continuous Learning Loop

Stages 3–4 of the Phased Rollout move from a few teams to many (~10–15 teams, ~30–50 people per wave). That is where the Delivery J-curve, the Verification Tax and shadow tooling hit hardest. Scaling works when every wave feeds a **continuous learning loop**: what we learn from telemetry becomes shared assets, guardrails and playbooks for the next wave.

The continuous learning loop (runs once per rollout wave, and continuously inside each team)

1 · ADOPT
Onboard the next wave on golden paths: catalog-approved agents, templates and durable skills. A champion in every value stream.
→
2 · MEASURE
Instrument the Dimensions SLIs from day one: active vs dormant users, FAR/FRR and rework, token cost and Verification Tax. Roll up via the OKR → Telemetry cascade.
→
3 · LEARN
Review signals by lens: weekly team retros (IC), monthly value-stream reviews (M), quarterly OKR resets (L). Find where the J-curve dip is deepest, and why.
→
4 · IMPROVE & CODIFY
Turn fixes into reusable assets: prompts, skills, guardrails, runbooks. Recalibrate agent autonomy (HITL ↔ HOTL) against error budgets.
→
5 · SCALE & SHARE
Promote proven local tools into shared capabilities: one platform, N value-stream instances. Publish playbooks; retire low-ROI workflows.

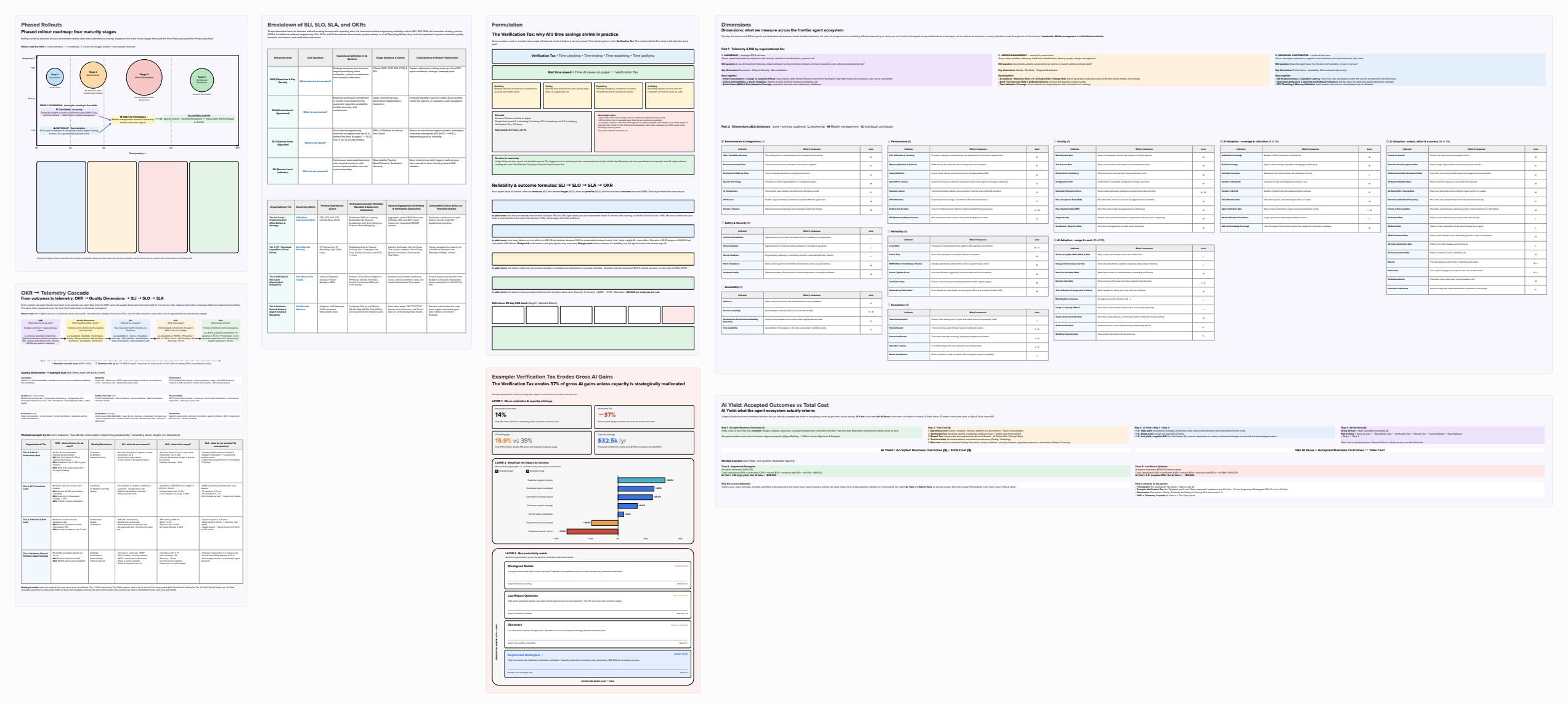
↺ **Feed back:** each wave's learnings become the starting kit for the next wave, so adoption gets faster and cheaper over time.

What Scale Demands

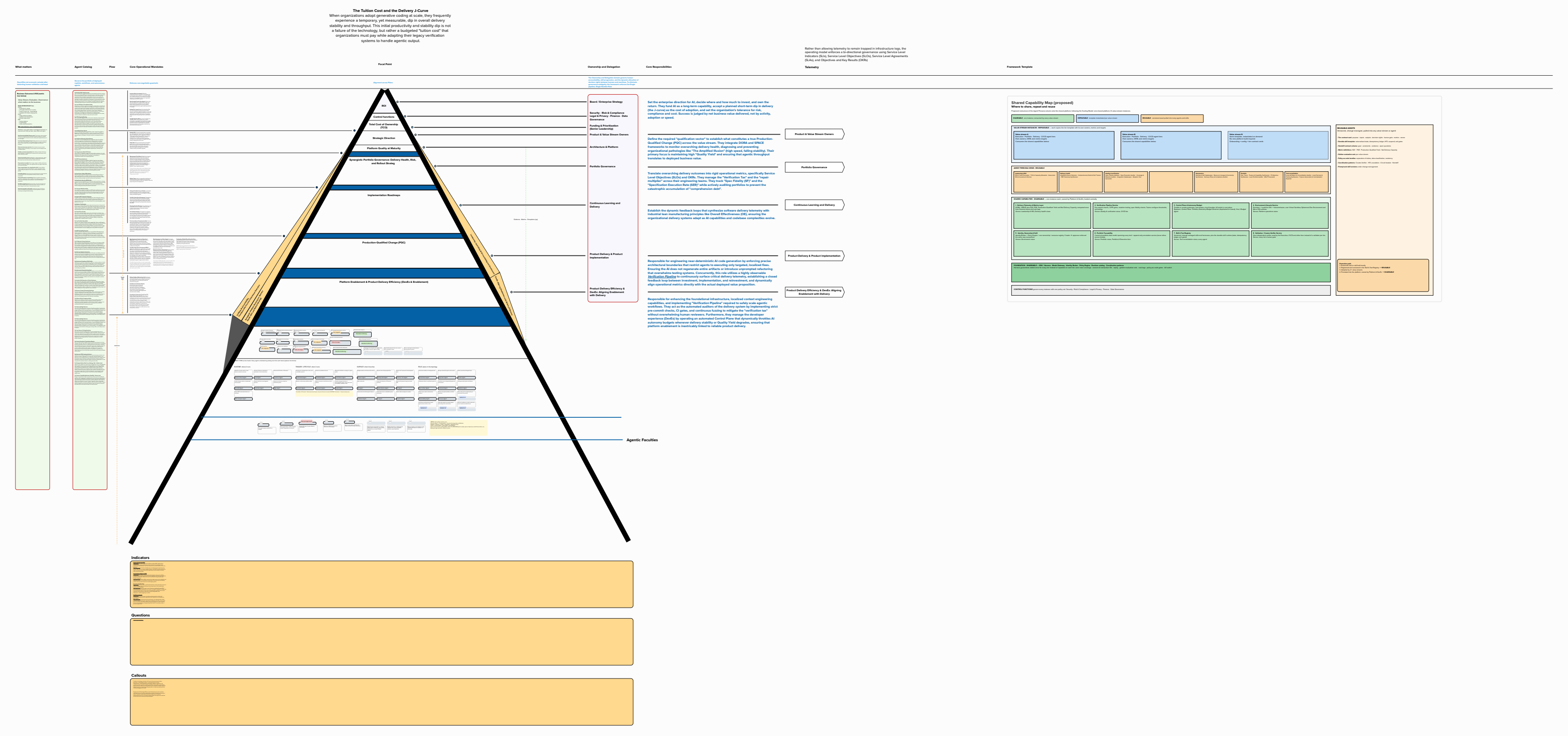
What scale demands : challenges to plan for, and the resources, tools and support that address them

	ADOPTION people use it, and keep using it	**ROLLOUT** expand to more teams safely	**INTEGRATION** fits real workflows and systems	**CONTINUOUS LEARNING** the loop gets faster each wave
CHALLENGES what breaks at scale	• Uneven uptake: power users vs dormant licenses (shelfware) • Trust gap: high rework and override/fallback push people back to manual • Product, QA and Ops left behind Dev (role coverage) • Change fatigue during the J-curve dip	• Delivery J-curve: temporary drop in stability and throughput as each wave onboards • Shadow AI and duplicate tools across value streams • Uneven guardrail and policy coverage per team • Token cost growing faster than value delivered	• Agents bolted onto workflows instead of designed into them • Fragmented toolchains, APIs and schema contracts • Context gaps: low repo AI-knowledge coverage • Verification Tax shifting load onto reviewers	• Telemetry exists but nobody acts on it • Local wins don't travel between teams • Metrics read out of context across tiers (or gamed) • Slow feedback from production defects back to prompts and skills
RESOURCES people · budget · time · content	• AI champions network (one per team) with protected time • Role-based learning paths (Dev, Product, QA, Ops) • Office hours and pairing sessions • License budget tied to active use, not headcount	• Platform / enablement team that owns the golden paths • Wave plan: ~10–15 teams, ~30–50 people per wave (Stage 3) • Tuition budget: planned capacity for the J-curve dip • Security and governance reviewers sized to each wave	• Integration engineers embedded in value streams • Reference architectures and API / schema contracts • Curated context: repo indexing, docs, domain knowledge packs • Time to redesign workflows, not just add tools	• Named owner for the Dimensions dashboard and analytics • Community of practice and internal showcases • Playbook library and durable-skill repository • Retro time built into the sprint cadence
TOOLS platform and tooling	• Agent Catalog as the single storefront for approved agents and skills • Low-latency in-IDE and in-browser assistants • Prompt, skill and template libraries • Self-serve onboarding and sandbox environments	• Shared platform with N value-stream instances • Policy-as-code guardrails, secret scanning, PQC gates • Feature flags and staged rollout per team • Cost controls: token budgets and model routing	• MCP servers / connectors to core systems • Agent harness with HITL / HOTL controls and an audit trail • AI-aware CI/CD: build, test and review gates • Observability: traces, SLIs and error budgets	• Dimensions dashboard by lens (L / M / IC) • Eval harness: output quality, FAR / FRR, instruction adherence • In-tool feedback capture (ratings, edits, overrides) • Experiment tracking: A/B tests of prompts, models and skills
SUPPORT operating model and governance	• Executive sponsorship plus middle-management scorecards (Stage 1 foundation) • Recognition for reuse and contribution • Clear acceptable-use policy and safe-to-try zones • Support desk with an SLA (response ≤ 30 min)	• Readiness criteria per wave: SLIs live, baselines captured • Error-budget rule: pause or roll back when breached • Portfolio review that halts scale-up of low-ROI workflows • A single named owner for every catalog agent	• Architecture review for agent-to-system access • Data-access and privacy approvals • Clear split of ownership: shared platform vs value stream • Incident process that covers AI-caused defects	• Loop cadence: weekly team · monthly value stream · quarterly OKR reset • Forum that promotes local tools into shared capabilities • Decision log: what changed, why, and its measured effect • Evidence-based autonomy recalibration (HITL ↔ HOTL)

The What: Measuring the frontier agent ecosystem

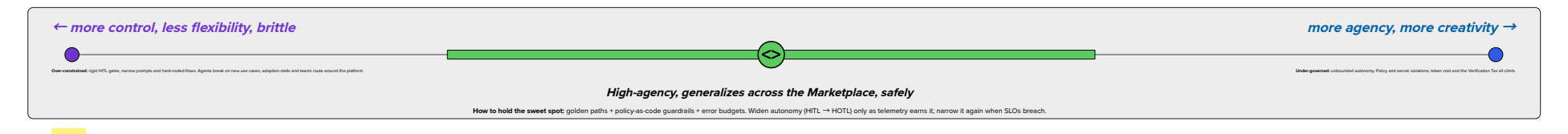


The How:



Agent Building

Agent Control: Deterministic vs Creative


← more control, less flexibility, brittle
more agency, more creativity →
<>

Over-constrained: rigid HITL gates, narrow prompts and hard-coded flows. Agents break on new use cases, adoption stalls and teams route around the platform.

Under-governed: unbounded autonomy. Policy and secret violations, token cost and the Verification Tax all climb.

High-agency, generalizes across the Marketplace, safely

How to hold the sweet spot: golden paths + policy-as-code guardrails + error budgets. Widen autonomy (HITL → HOTL) only as telemetry earns it; narrow it again when SLOs breach.

Offline Human Alignment & Analysis

Humans label the signals, so the feedback loop learns the right thing

① Analysis & labeling pipeline (offline)

Agent Signals
Traces, decisions, user feedback and SLI breaches, sampled for human review
→
Human Labelers
Rubric-anchored grading; calibrated raters, agreement checked, disagreements adjudicated
→
Ground Truth (Labeled Dataset)
Versioned golden set; a held-out slice is never used for tuning
→
Signal Analysis
Error taxonomy, label drift, and where agent and human judgment disagree

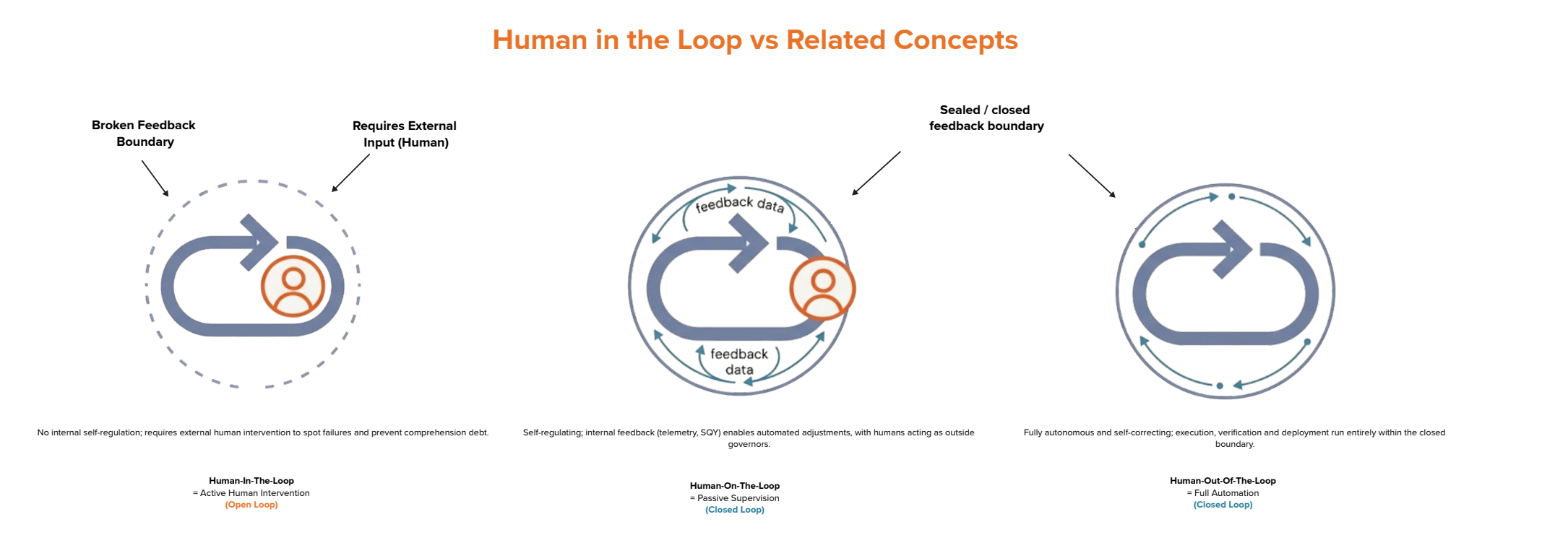
② Evaluate → tune → ship

Agent Decision
e.g. routing: scores the input → enhance / keep
→
Evaluate
Agent decision vs. golden-set truth → precision & recall
→
Tune
Prompts, thresholds, tools. Raise the bar; bias toward recall where misses are costly
→
Ship
Deploy once precision & recall clear the bar; new signals flow back to labeling

Loop: re-sample unseen golden-set items

Why it matters: the feedback loop is only as good as its labels. If signals are mislabeled, the agent is tuned toward the wrong target. Human alignment and label QA are part of agent building, not an afterthought.

HITL vs HOTL: Open Loop vs Closed Loop


Human in the Loop vs Related Concepts
Broken Feedback Boundary
Requires External Input (Human)
Sealed / closed feedback boundary
feedback data
feedback data

No internal self-regulation; requires external human intervention to spot failures and prevent comprehension debt.

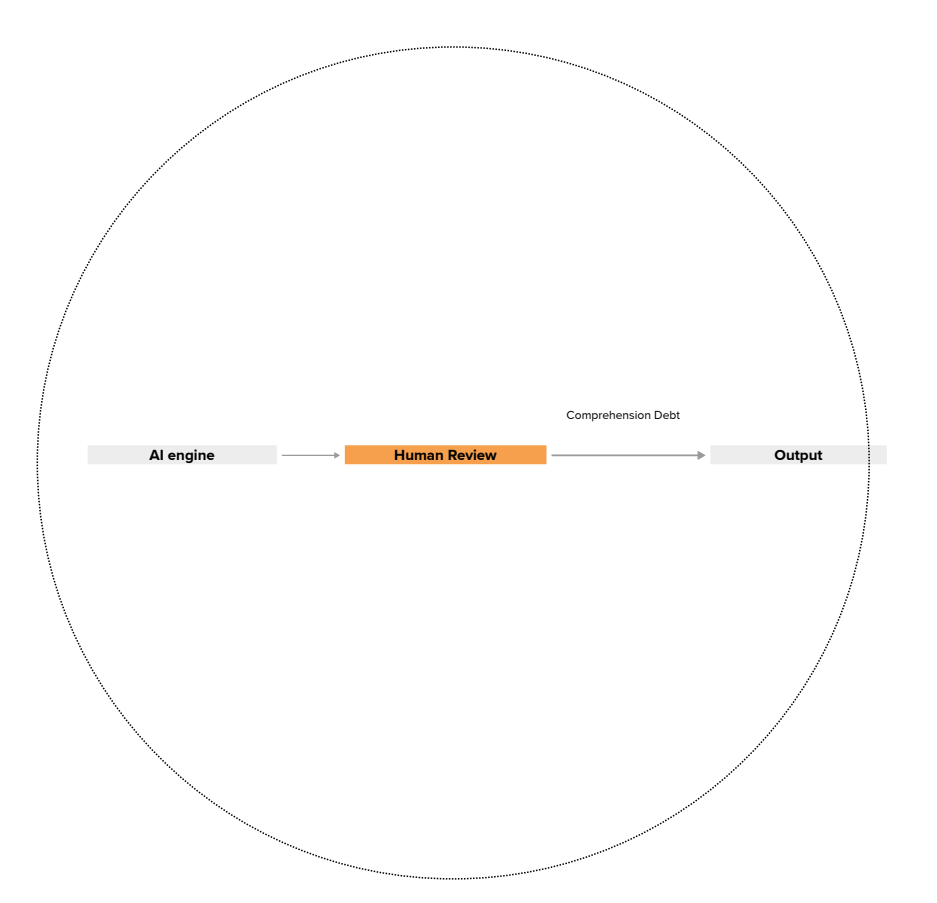
Human-In-The-Loop
= Active Human Intervention
(Open Loop)

Self-regulating; internal feedback (telemetry, SQY) enables automated adjustments, with humans acting as outside governors.

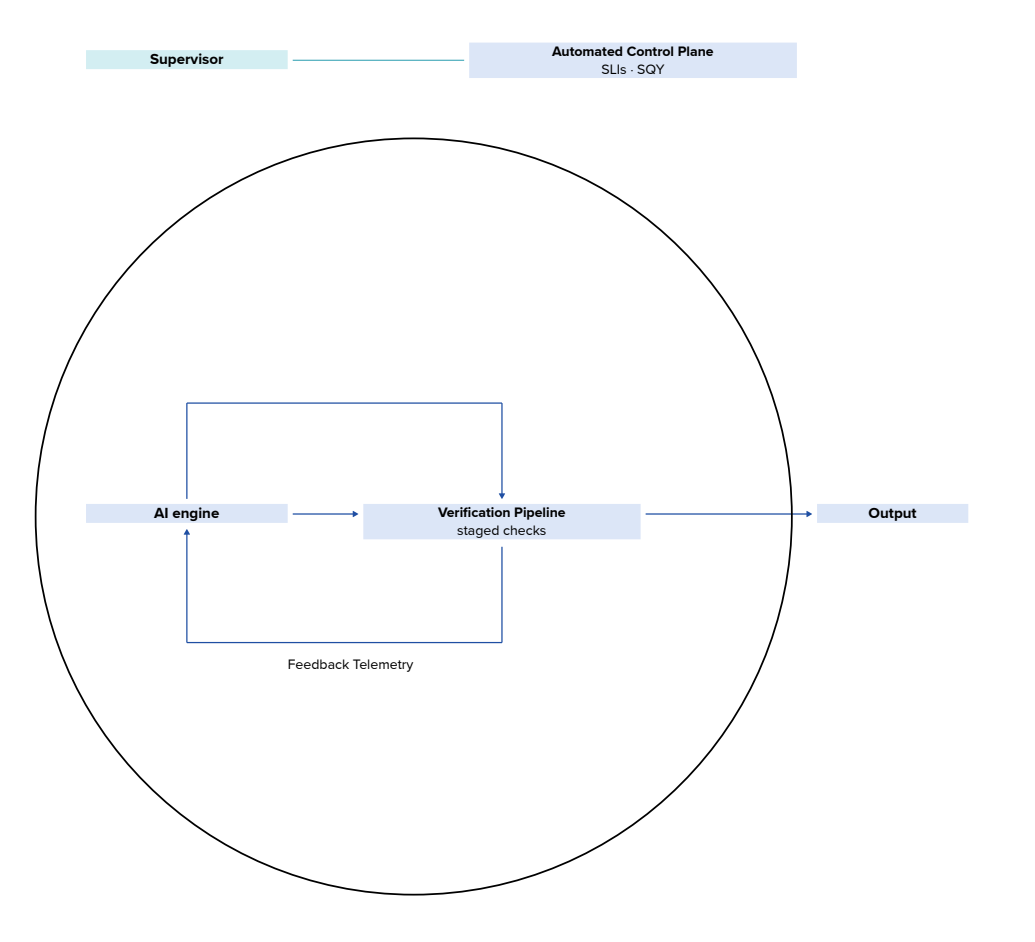
Human-On-The-Loop
= Passive Supervision
(Closed Loop)

Fully autonomous and self-correcting; execution, verification and deployment run entirely within the closed boundary.

Human-Out-Of-The-Loop
= Full Automation
(Closed Loop)



Open Loop (HITL)



Closed Loop (HOTL)

Agent Lifecycle Evaluation

Every agent is evaluated on AI Yield at every stage of its life, not just at launch

1 · Design & Baseline
Define the job, the owner, the autonomy level (HITL → HOTL) and the AI Yield target. Build the golden set and record the pre-agent baseline cost.
Gate: success criteria and target signed off
→
2 · Offline Evaluation
Run against the human-labelled golden set: precision & recall, rubric scores, safety tests. Estimate the Step 2 costs.
Gate: clears the bar on unseen golden-set items
→
3 · Pilot (HITL)
Limited users; a human reviews every output. Measure Accepted Outcomes, Verification Tax and Rework Tax on real work.
Gate: pilot AI Yield ⩾ 1.0, zero guardrail violations
→
4 · Production (HOTL)
Scale out under telemetry: SLIs, SLOs and error budgets. Humans supervise from outside; sampled outputs go back to labelling.
Gate: SLOs held, error budget not exhausted
→
5 · Periodic Review
Monthly or quarterly, per agent: AI Yield = Accepted Outcomes ÷ Total Cost; Net AI Value = Outcomes − Cost. Check drift and re-sample unseen golden-set items.
→
6 · Decide
Every review ends in one of three outcomes: keep the agent running, release a new version, or retire it.

Keep running
AI Yield > 1.0 and SLOs met → stays in Production until the next review

Keep: back to Production until the next review

New version (vN+1)
AI Yield < 1.0 but fixable, or a model, tool or policy change → changes introduced, re-evaluated from the top

New version (vN+1): after evaluation, changes are introduced and the agent goes back to the top (stage 1)

Retire the agent
No longer needed, superseded, or AI Yield < 1.0 with no fix → switched off and no longer used; archived with its prompts, evals, telemetry and audit trail

Measured at every stage (from AI Yield): Step 1 Accepted Business Outcomes ($) · Step 2 Total Cost (operational, Verification Tax, Rework Tax, technical debt, risk) · Step 3 AI Yield = Step 1 ÷ Step 2 · Step 4 Net AI Value = Step 1 − Step 2

Rules: each new version starts again at stage 1 and earns its autonomy back · AI Yield is tracked per version, so versions can be compared · retired agents are archived, not deleted